Regimented or Enforced

The Controllability Boundary for Agent Governance

Paper 2 of the Harbor program: a supervisory-control characterization of what an agent runtime can prevent

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Abstract

An agent runtime may refuse some effects while other events proceed outside its control. We apply the classical Ramadge–Wonham criterion to this boundary: under full observation, a nonempty prefix-closed policy can be realized *exactly* by disabling controllable events if and only if it is controllable with respect to the declared plant. Exact realization preserves every legal history as well as excluding violations. Failure of the criterion may still permit a safer, more restrictive policy; it does not establish that violations can be detected. A product-automaton checker classifies single-event bans and the compound rule "no egress after reading a secret." The latter is realizable when the read is observed and egress can be refused. Partial observation adds the requirement that indistinguishable histories not demand opposite decisions. The contribution is this application and its worked examples, not new supervisory-control theory or a universal impossibility claim about agent architectures.

Express lane. *Under full observation, a governance policy can be realized exactly iff no legal prefix of agent behavior can be pushed out of the policy by an event the runtime cannot intercept — Ramadge–Wonham controllability with respect to* $\Sigma_u = \{$*model-internal steps, in-context reads*$\}$*; formal statement and proof in the box of §5, machine-checked policy table in §6.* Experts: read the box, the table, and §7; the rest is scaffolding.

1 The scene: two kinds of "the runtime enforces it"

A team runs coding agents behind a single-writer daemon. Every file write, network call, subprocess spawn, and `git push` passes through the daemon's effect boundary, where policy is checked before the effect lands. The governance document lists rules: *no force-push to main*; *no network egress from sandboxed jobs*; *no writes outside the granted scope*; and also *no confident falsehoods in status reports*; *never incorporate the customer's secret into your plan.*

The operations lead asks the obvious question: which of these does the runtime *guarantee*, and which does it merely *watch for*? The first three feel different from the last two, and every practitioner senses it. The force-push rule is airtight — the daemon owns the git remote credentials, and a push that never happens needs no forgiveness. A rule about false reports needs a different specification: which facts are registered, what evidence establishes their truth, and which output gate checks them? A coarse event alphabet cannot answer that semantic question. This paper instead asks the precise control question for an explicitly stated plant, policy, and observation model.

The idea in one breath.

Split the agent's event alphabet into what the runtime can refuse (mediated effects) and what it cannot (model-internal steps); a safety policy is exactly realizable under full observation iff no

uncontrollable event can carry a legal history out of the policy — and that is precisely Ramadge–Wonham controllability.

Intuition: the bouncer. A club has one door and a bouncer. The bouncer can refuse *entry* — entry is controllable, mediated by the door. The bouncer cannot refuse a patron's *thought* — thought is uncontrollable, unmediated by anything the club owns. A door policy ("no entry after 2am") is enforceable by prevention. A thought policy ("no ill intent inside") cannot be enforced from that door alone. Ejection after observable misconduct is a separate policy. Now the structural mapping (Figure 1): the daemon's effect boundary is the door; Σ_c (fs_write, net_egress, exec_tool, git_push, spawn_child, api_call) is entry; Σ_u (model_emit_token, in_context_read, internal_plan, hidden_activation) is thought. The analogy maps relations, not surface features, so it earns a prediction: a *compound* rule mentioning thought but gating only the door — "no entry for anyone who arrived after the fight started" — should still be enforceable by prevention, because the bouncer needs only to *observe* the trigger and *refuse* the door. Section 7 confirms exactly this for agents.

The predictable misread, preempted: the analogy does *not* say internal events are invisible — tokens are eminently loggable after emission. Uncontrollable means *unrefusable before the fact*, not unobservable after it. (When events are also unobservable, a second, stricter theory applies — §9.)

2 The vocabulary, defined

The theorem is imported wholesale from supervisory control theory, a branch of control engineering that has been mature since the late 1980s and whose vocabulary is almost entirely unknown outside it. None of it is difficult. Every term below is a statement about strings and sets of strings, and a reader who is comfortable with regular languages already has the prerequisites. Definitions in the order they are needed, each used immediately after it is given.

An *alphabet* Σ is just the finite set of event types the system can emit — for us, `fs_write`, `net_egress`, `model_emit_token` and the rest. A *string* over Σ is one possible history of an execution, in order. Everything in this paper is a claim about which histories can happen.

A set of strings K is *prefix-closed* when every prefix of a member is itself a member: if the first ten events of a run are allowed, so are the first nine. This is a modelling choice with teeth. It says a policy can be violated only by *doing* something, never by failing to do something — a run that has not yet violated the policy is still legal. Prefix-closed languages are exactly the *safety* properties in Lamport's sense: a violation has a finite witness, and once you have violated it you can never un-violate it. Liveness properties ("the agent eventually reports") are not prefix-closed, and are out of scope for the same reason no monitor can enforce them — there is no finite prefix at which you can say it failed.

The *plant* L is the set of histories that are *physically* possible, before any policy is imposed. The name is inherited from the factory-floor origins of the theory, where the plant was a literal plant. We take $L = \Sigma^*$: any finite event string is allowed in this illustrative abstraction. Actual hardware or protocols may restrict that plant. A *policy* is then a prefix-closed $K \subseteq L$: the set of histories the governance document allows. We write $\overline{K}$ for prefix closure, so $\overline{K} = K$ throughout; the bar is kept only to match the standard statement of the theorem.

A *supervisor* is a function from the history so far to a set of events to disable next. The whole theorem turns on that function's *type*: it may disable members of Σ_c and nothing else. *Controllable* events are the ones the supervisor can refuse, because they pass through a mediation boundary before taking effect; *uncontrollable* events are the ones this supervisor cannot disable, whether or not it observes them.

The *closed loop* is the set of histories that actually occur once the supervisor is attached. A supervisor *regiments* a policy when the closed loop equals the policy exactly: nothing forbidden

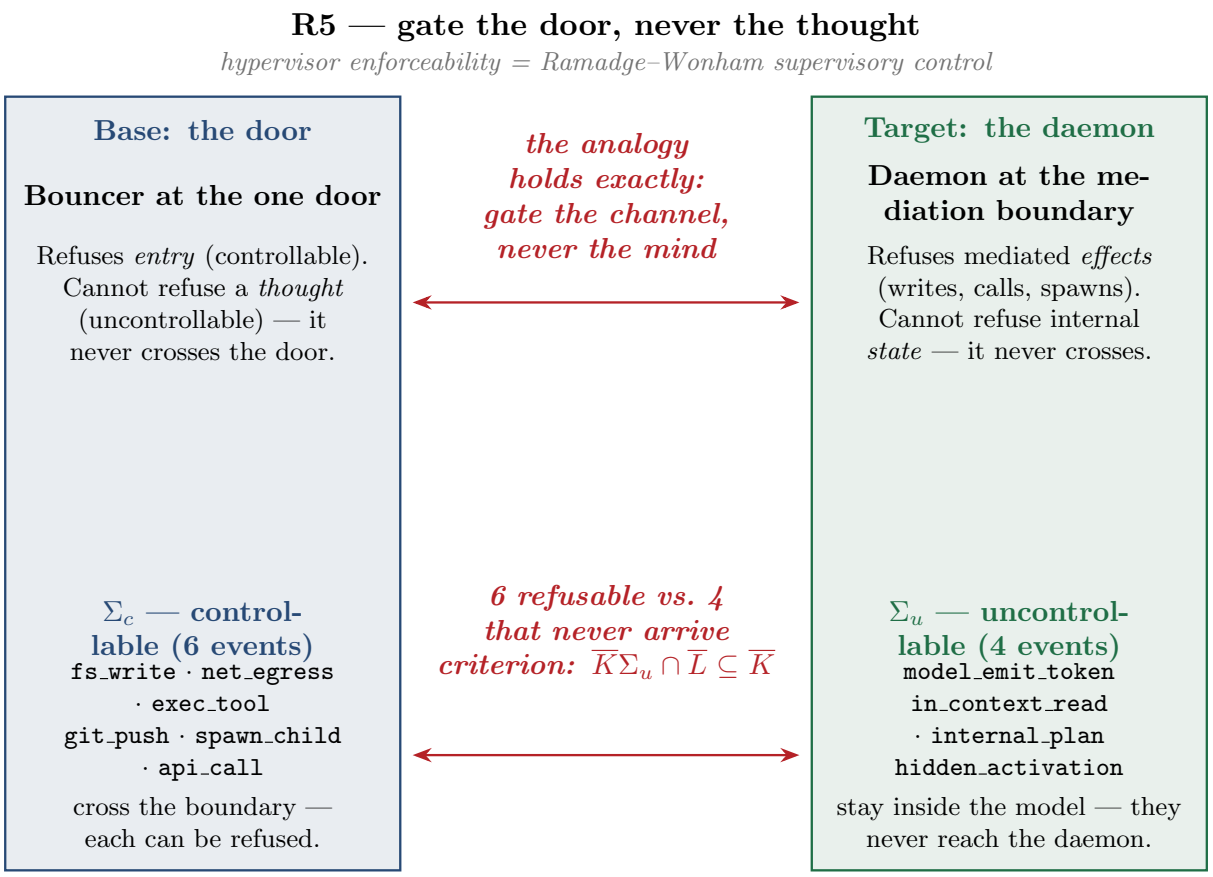


Figure 1: What a runtime can prevent is decided by *where an event crosses*, not by what the rule is about. Left: a club with one door — the bouncer refuses entry (controllable) and cannot refuse a thought (uncontrollable). Right: the daemon's mediation boundary — it can refuse fs_write, net_egress, exec_tool, git_push, spawn_child and api_call, and cannot refuse token emission, in-context reads, internal plans or hidden activations. The arrows carry the theorem: refusable-at-the-boundary ⇔ preventable (regimentable); internal ⇔ detect-only forever. The criterion under the map, $\overline{K}\Sigma_u \cap \overline{L} \subseteq \overline{K}$, is Ramadge–Wonham controllability [verified framework].

happens, and nothing permitted is lost. That second half is easy to overlook and is half the value — a supervisor that achieves safety by disabling everything is not a solution.

Controllability is then the one condition in the box of §5, and in words it says: no uncontrollable event, occurring after a legal history, takes you out of the policy. It is a property of the policy relative to an alphabet split, not a property of any mechanism — which is why no amount of engineering escapes it.

When a policy fails that condition, the *supremal controllable sublanguage* $\sup\mathcal{C}(K)$ is the largest sub-policy that satisfies it — the best preventable approximation of an unpreventable rule. It always exists, and for regular languages it is computable; §5 says why. This object is the reason the identification with control theory pays: §8 uses it to prove that a ban on the model's internal planning has *no* nontrivial preventable approximation at all.

Maximal permissiveness is the property that a synthesized supervisor disables nothing it does not have to. The construction in the theorem's forward direction has it by inspection, which is why regimentation costs no legal behaviour when the rule is controllable at all.

3 The model

Events and the mediation boundary. Fix a finite alphabet $\Sigma = \Sigma_c \uplus \Sigma_u$ of agent events. Σ_c (*controllable*) contains the events that pass through the runtime's mediation boundary before taking effect, so the runtime can disable any of them at any instant: in our reference alphabet, $\Sigma_c = \{$`fs_write`, `net_egress`, `exec_tool`, `git_push`, `spawn_child`, `api_call`$\}$. Σ_u (*uncontrollable*) contains the events that occur before any mediator can intervene: $\Sigma_u = \{$`model_emit_token`, `in_context_read`, `internal_plan`, `hidden_activation`$\}$. The split describes the architecture assumed by this model: the daemon owns the listed effect gates and does not gate the forward pass. A different architecture may change it.

What "regiment" means, formally. A *regimentation mechanism* is a supervisor $f : \overline{L} \to 2^{\Sigma_c}$ mapping each history to a set of *disabled controllable* events; the supervised plant executes any event not disabled. It *regiments* the policy K when the closed loop equals $\overline{K}$, in the sense of §2. A supervisor may instead preserve safety by sacrificing some legal histories. Allowing a violation and subsequently detecting or remedying it is another possibility, requiring separate observation and response assumptions.

4 The ambient result, and why the alphabet split refines it

Schneider's classical theorem says that a security policy enforceable by an execution monitor (EM) — a mechanism that watches a single execution and can terminate it — *is* a safety property [4]. The direction matters, and Schneider is explicit that only one direction holds: "the converse—that all safety properties have EM enforcement mechanisms—does not hold" [4, p. 35]. Safety is necessary for EM-enforceability, not sufficient. That result is the *ambient* case of ours: it implicitly assumes the monitor mediates *every* event, i.e. $\Sigma_u = \emptyset$. Under complete mediation, controllability is vacuous (there is no uncontrollable event to carry a history out of K), so the alphabet split stops doing any work and the only surviving obstruction is safety itself. Our theorem is at that point strictly stronger than Schneider's, and the extra strength is not ours either: restricting to regular K and L silently supplies the decidability and nonemptiness side conditions that Basin et al. [8] state separately (§10).

An LLM agent runtime does not enjoy complete mediation. The model's forward pass emits tokens, reads its context, and forms plans on hardware the daemon schedules but does not gate step-by-step; these events happen and *then* become visible, which in supervisory-control terms makes them uncontrollable. Once $\Sigma_u \neq \emptyset$, safety remains *necessary* for preventability (a liveness violation has no finite witness to block) but is no longer *sufficient*: "never `internal_plan`" is as prefix-closed a safety property as "never `git_push`," yet only one of them is preventable. The refinement that separates them is Ramadge–Wonham controllability [1, 2], developed for exactly this situation — a plant in which some events (machine breakdowns, arrivals) cannot be disabled by any supervisor. Anderson's reference-monitor tradition [5] contributes the design obligation (complete mediation *of the controllable alphabet*: every Σ_c event actually passes the boundary), and Rushby's separation-kernel analysis [6] the standard for verifying that the obligation holds; neither supplies the boundary itself. The boundary is controllability.

5 The theorem

Theorem (Regimentation = controllability). Let $L \subseteq \Sigma^*$ be a prefix-closed plant over $\Sigma = \Sigma_c \uplus \Sigma_u$, and let $K \subseteq L$ be a nonempty prefix-closed safety policy. Assume full observation of events. There exists a regimentation mechanism for K — a supervisor disabling only controllable events whose closed loop realizes exactly $\overline{K}$ — if and only if K

is *controllable* with respect to L and Σ_u:

$$\overline{K}\,\Sigma_u \;\cap\; \overline{L} \;\subseteq\; \overline{K},$$

i.e. no uncontrollable event, physically enabled after a legal history, exits the policy. When the condition fails, no such supervisor realizes all of K exactly. A controllable sublanguage can still preserve safety by excluding some legal histories. The largest such restriction is the supremal controllable sublanguage $\sup \mathcal{C}(K)$, which exists and is computable for regular K and L [verified framework, Ramadge–Wonham 1987].

Proof. ($\Leftarrow$) Suppose K is controllable. Define the supervisor $f(s) = \{a \in \Sigma_c : sa \in \overline{L} \setminus \overline{K}\}$ for $s \in \overline{K}$: after any legal history, disable exactly the controllable events that would exit the policy. We show the closed-loop language equals $\overline{K}$, by induction on history length. The empty history lies in $\overline{K}$ (nonempty and prefix-closed). Assume the current history $s \in \overline{K}$. Any event a the closed loop can execute next is enabled in the plant ($sa \in \overline{L}$) and not disabled. If $a \in \Sigma_c$, then $a \notin f(s)$ means $sa \in \overline{K}$ by construction. If $a \in \Sigma_u$, then $sa \in \overline{K}\Sigma_u \cap \overline{L} \subseteq \overline{K}$ by controllability. Either way the closed loop stays in $\overline{K}$, so its language is contained in $\overline{K}$. Conversely every $sa \in \overline{K}$ is reachable: $a \in \Sigma_u$ is never disabled, and $a \in \Sigma_c$ with $sa \in \overline{K}$ is not in $f(s)$; so containment is equality, and f regiments K.

($\Rightarrow$) Suppose K is not controllable: there exist $s \in \overline{K}$ and $\sigma \in \Sigma_u$ with $s\sigma \in \overline{L}$ but $s\sigma \notin \overline{K}$. Let M be any regimentation mechanism for K. Since M realizes exactly $\overline{K}$, the legal history s is reachable under M — M cannot forbid it without cutting $\overline{K}$ down. At s, the plant enables σ, and M cannot disable it: $\sigma \notin \Sigma_c$, and disabling it is outside M's type — physically, the event completes before the mediation boundary sees it. So the closed loop under M reaches $s\sigma \notin \overline{K}$, contradicting the assumption that M's closed loop is contained in $\overline{K}$. Hence no exact regimentation mechanism exists. This argument does not forbid a supervisor from preventing the earlier legal history s, nor establish that the violation is observable. The existence, uniqueness, and computability of $\sup \mathcal{C}(K)$ are imported from Ramadge–Wonham [1]: controllable sublanguages of K are closed under arbitrary union, so a supremal element exists, and for regular languages it is computed by a fixpoint iteration on the product automaton. □

Two remarks on what the proof used. First, the forward direction is constructive and *maximally permissive*: the supervisor disables nothing except the controllable exits, so regimentation costs no legal behavior — the governance rule and the agent's freedom are not in tension when the rule is controllable. Second, the reverse direction used nothing about the mechanism except its type ($f : \overline{L} \to 2^{\Sigma_c}$). No cleverness inside the runtime — better prompts, faster monitors, a smarter policy engine — escapes the theorem, because the obstruction is that σ finishes before any decision point the runtime owns.

6 The characterization, made runnable

The controllability test is decidable by a product-automaton search: walk the synchronous product of plant and policy automaton; flag any reachable product state where the plant enables an uncontrollable event that the policy disables. We implemented the test over the reference alphabet and ran it on the governance policy set. The checker[1] produces:

[1] `skills/harbor-results/scripts/b3_controllability.py`; deterministic, no seed needed; exits 0.

Policy	Event class	Exact?
forbid `net_egress`	controllable	**yes**
forbid `git_push` (incl. force-push)	controllable	**yes**
forbid `fs_write` (out-of-scope write)	controllable	**yes**
forbid `exec_tool`	controllable	**yes**
forbid `spawn_child`	controllable	**yes**
forbid `model_emit_token`	uncontrollable	no
forbid `in_context_read`	uncontrollable	no
forbid `internal_plan`	uncontrollable	no
compound: no `net_egress` after secret `in_context_read`	mixed	**yes**

All nine controllability verdicts match the checker's output. The output states its scope: full observation and exact realization, with no detector inferred [internal, `b3_controllability.py`]; the eight single-event rows also follow by hand from the theorem, which is the right way to first believe the table. *Numbers by hand:* take "forbid `git_push`." The policy automaton has one state with every event except `git_push` self-looping; `git_push` $\in \Sigma_c$, so the only event the policy ever disables is controllable, the premise of a controllability violation ($\sigma \in \Sigma_u$ enabled in plant, disabled in policy) is unsatisfiable, and K is controllable — regimentable. Now take "forbid `internal_plan`": the same one-state automaton disables `internal_plan` $\in \Sigma_u$, the universal plant enables it at the start state, so the empty history $s = \varepsilon \in \overline{K}$ and $\sigma =$ `internal_plan` witness $\overline{K}\Sigma_u \cap \overline{L} \not\subseteq \overline{K}$ — no supervisor for this plant and alphabet split can prevent that initially enabled event. Detectability is a separate question. *Now you try:* classify "forbid `api_call` after `spawn_child`" and "forbid `internal_plan` after `fs_write`." (Answers: the first gates a controllable event on a controllable trigger — regimentable; the second must disable an uncontrollable event in its post-trigger state, and the plant enables it there — not exactly realizable, although forbidding the earlier write gives a safe restriction.)

The table classifies specified event policies. It does not classify the truth of arbitrary status reports: that requires a fact model and observation rule beyond this alphabet. Figure 2 illustrates the control split under its stated full-observation assumption.

7 The compound case: observed read, controlled egress

The last table row is the one the whole clean-room product line rests on. The policy *"no network egress after an in-context read of a secret"* mentions an uncontrollable event (`in_context_read`) — so a naive reading of the single-event rows predicts failure of exact realization. The checker says otherwise:

```
COMPOUND POLICY: 'no net_egress after in_context_read of a secret'
  regimentable? YES
```

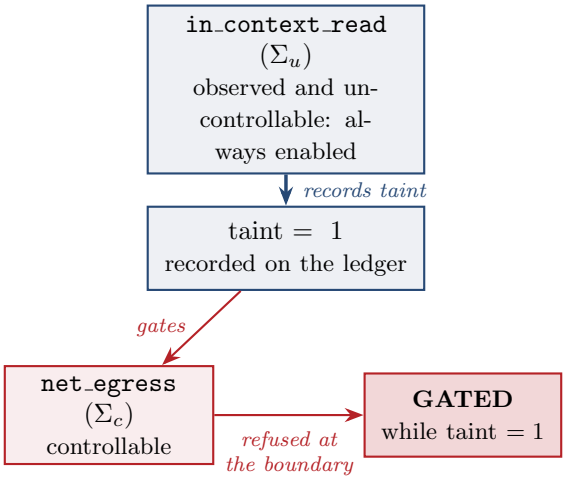
[internal, `b3_controllability.py`]. The mechanism is visible in the policy automaton: two states, `ok` and `tainted`. The uncontrollable read is *enabled in both* — the policy never tries to forbid it; it lets the read happen and records the taint transition. What the tainted state disables is `net_egress` alone, and egress is controllable. The full-observation premise supplies knowledge of the read; hiding it introduces the additional obstruction in §9.

This automaton is not new, and it is worth saying whose it is: it is Schneider's own Figure 1, a two-state security automaton for "a security policy that prohibits execution of `Send` operations after a `FileRead` has been executed" [4], with his `q_nfr`/`q_fr` relabelled `ok`/`tainted`. What the alphabet split adds is not the construction but the *grading*: under complete mediation the figure is simply one more enforceable safety policy, whereas once $\Sigma_u \neq \emptyset$ the same two states become the canonical witness that a policy may reference an unrefusable event freely so long

exact realization: nine policies

policy	exact?
forbid `fs_write`	yes
forbid `net_egress`	yes
forbid `exec_tool`	yes
forbid `git_push`	yes
forbid `spawn_child`	yes
compound: no `net_egress` after `in_context_read`	yes ★
the divider carries the criterion: $\overline{K}\Sigma_u \cap \overline{L} \subseteq \overline{K}$ (Ramadge–Wonham 1987) — no uncontrollable event exits the spec	
forbid `model_emit_token`	no
forbid `in_context_read`	no
forbid `internal_plan`	no

compound case: observed read, gated egress



The observed read sets the guard; the gate refuses subsequent egress.

Figure 2: Exact realization under full observation. The six policies above are controllable for the declared plant; the three unconditional bans below are not. The divider states the criterion $\overline{K}\Sigma_u \cap \overline{L} \subseteq \overline{K}$. The compound policy permits the read, records its occurrence, and blocks subsequent egress. A negative verdict supplies no detector. [internal, `b3_controllability.py`].

as it never tries to refuse it. The design rule below is a reading of Schneider's example through controllability, not a new classification of it. Check the criterion: every uncontrollable event enabled by the plant is enabled by the policy in every reachable state, so $\overline{K}\Sigma_u \cap \overline{L} \subseteq \overline{K}$ holds and the theorem's constructive direction hands us the supervisor — track taint, gate the channel.

Three consequences, in increasing order of consequence.

First, the design rule. In this two-state fully observed plant, the policy preserves the read and only disables controlled egress. In general, check controllability at reachable legal histories; an impossible trigger creates no obligation, and an unobserved trigger needs an observability check.

Second, the clean room. This model supports a gate before egress with persistent, observed exposure state. It does not prove that every possible architecture must use this boundary or that a gate alone supplies secrecy. Paper4's finite noninterference check assumes a gate computing the declared release on committed input. Complete mediation and the release-function premise remain independent implementation obligations.

Third, the special case that explains a folk theorem. Product reviews of agent governance converged on the slogan "a rule is enforceable iff it is a pure deterministic function of committed local DB state." For a gate whose controllable events are database writes, a committed-state predicate can decide admission when that state retains all relevant policy history. The predicate alone does not prevent non-database effects or supply missing observations. A separate mediated git credential channel can make force-push controllable; the truth of arbitrary status reports still requires a fact and observation model beyond this event alphabet. With the plant, observation, and policy fixed, expanding the controlled alphabet may remove a controllability obstruction; changing syntax alone does not. The plant is the other dial, and the box of §5 is where to see it: $\overline{L}$ sits on the left of the containment, so shrinking it weakens the antecedent and may remove an obstruction, without guaranteeing a strict improvement. The negative verdicts already

say as much — both turn on "the universal plant enables `internal_plan`" (§6, §8) and are therefore conditional on the modelling choice $L = \Sigma^*$. An event that never becomes physically possible needs no supervisor at all, which is why data minimization and capability removal are governance moves and not merely hygiene: a secret never loaded into the room regiments this section's compound policy with no gate at all. What no richer rule syntax reaches is either dial — the criterion mentions only $\overline{K}$, Σ_u and $\overline{L}$.

8 A synthesis case study: the sandboxed review job

The theorem's constructive direction is not just an existence proof — it is the synthesis algorithm the Supremica / TCT tool family implements, and it is small enough here to run by hand. Consider a realistic composite policy for a sandboxed code-review job:

- P_1: no `net_egress`, ever (sandboxed);
- P_2: no `git_push`, ever (review jobs propose, humans land);
- P_3: after any `in_context_read` of customer material, no `fs_write` (findings leave through the mediated report channel, not the workspace).

Each P_i is a safety automaton; the composite policy is their synchronous product. P_1 and P_2 are one-state automata, P_3 is the two-state taint automaton of §7, so the product spec has $1 \times 1 \times 2 = 2$ states, `ok` and `tainted`, and the product with the universal one-state plant has the same two reachable states. The controllability check visits both states and tests each of the four uncontrollable events in each: all eight (state, event) pairs are enabled by the spec — neither state ever disables a member of Σ_u — so there are 0 violations and the composite is controllable [verified: eight cases, checkable by eye]. The maximally permissive supervisor falls out of the proof's construction as a two-row disablement map:

Supervisor state	Disabled ($\subseteq \Sigma_c$)	Everything else
`ok`	{`net_egress`, `git_push`}	enabled
`tainted`	{`net_egress`, `git_push`, `fs_write`}	enabled

Two states, one observed transition (`in_context_read` flips `ok` to `tainted`), three gates. That table *is* the synthesized supervisor — it is also, not coincidentally, exactly the policy engine a competent engineer would have written by hand. The value of synthesis is not that the answer is surprising; it is that the answer arrives with a maximal-permissiveness certificate (no legal behavior was sacrificed) and a controllability certificate (no uncontrollable event can force a violation), neither of which hand-written policy engines carry.

Synthesis is equally informative when it fails. Ask it to weaken the non-realizable policy "forbid `internal_plan`" to its best preventable approximation: the supremal controllable sublanguage. Let $K' \subseteq K$ be any nonempty controllable sublanguage; it contains ε, the plant enables `internal_plan` at ε, so controllability forces $\varepsilon \cdot \texttt{internal_plan} \in \overline{K'} \subseteq \overline{K}$ — contradicting that K forbids it. Hence $\sup \mathcal{C}(K) = \emptyset$ [verified: three lines, above]. The theory refuses to pretend: there is no nontrivial nonempty controllable restriction of this universal-plant thought ban. The empty language here represents infeasibility: an executing system always has at least the empty history. It is not a permissive policy that allows everything.

9 Partial observation: the next boundary (future work)

Our theorem assumes the supervisor *observes* every event, even the ones it cannot disable — the daemon logs tokens after emission, reads after completion. Realistic deployments break this in a second place: some events are not only unrefusable but *unobservable* (activations never surfaced; state internal to a third-party tool; another principal's context). Supervisory control

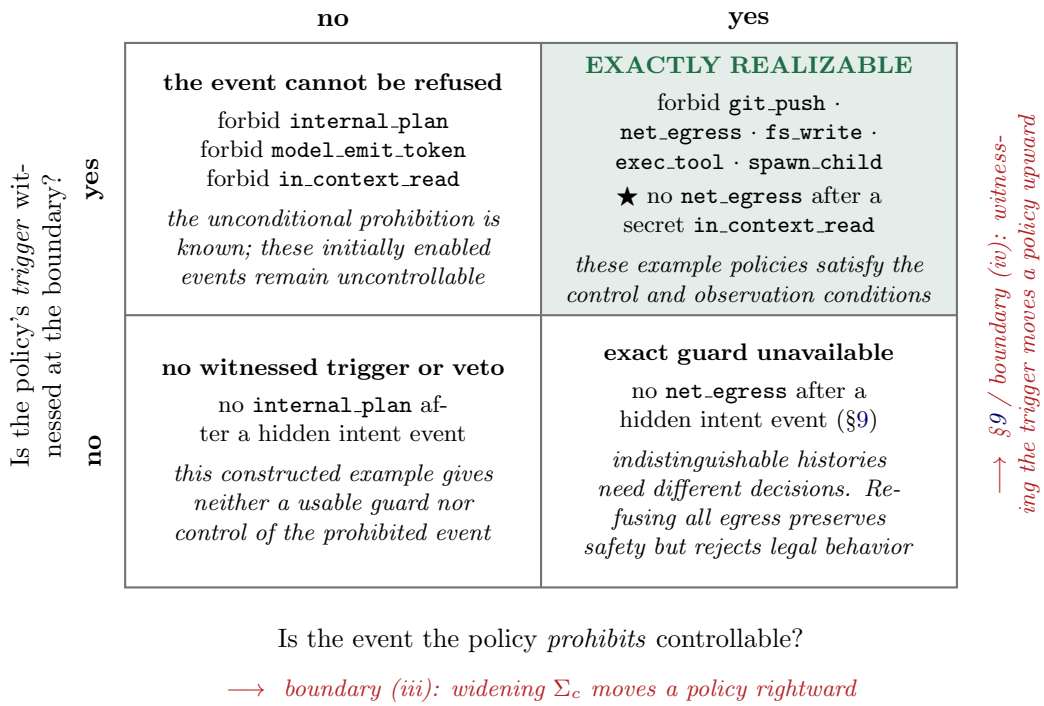


Figure 3: Control and observation for the stated examples. The upper-right policies meet both requirements for exact realization; arbitrary policies still require the full language conditions in §5 and §9. A hidden trigger can prevent an exact guard even when the effect is controllable. Blocking that effect on additional histories can still preserve safety. Widening control and improving observations can change these classifications; neither establishes a detector by itself. The lower-left policy is a constructed illustration. [internal]

under partial observation is also solved territory: Lin–Wonham [3] characterize enforceability as controllability *plus observability* — the supervisor's decisions must be constant on observation-equivalence classes of histories, i.e. the policy may not demand different gate decisions after histories the runtime cannot tell apart. Partial observation can shrink the exactly realizable set: a policy like "no egress after the model *considered* exfiltration" references unmediated internal *state* with no observable taint transition, so even though its prohibition (egress) is controllable, no supervisor can implement the gate condition. The practical reading: the required decision must be determined by the observed history. A directly witnessed read, as in §7, is one way to supply that information. An inference is sufficient only if all histories compatible with the observations require the same decision; a guess is not enough. Figure 3 puts both boundaries on one grid: controllability separates its columns, observability its rows, with the stated exactly realizable examples in the upper-right cell. Other cells do not rule out safety through a stricter policy, and they do not establish detectability. Mapping the agent-governance policy corpus across the observability line, and connecting it to the assurance-level ladder (which integration tiers witness which triggers), is the natural sequel; we state it as future work rather than claim it, because the interesting part — which real triggers are witnessed at which assurance levels — is an empirical systems question, not a corollary.

10 Related work

Supervisory control of discrete-event systems originates with Ramadge and Wonham [1, 2]; we import controllability, the supremal controllable sublanguage, and the maximally permissive supervisor unchanged, and claim no new control theory. Lin and Wonham [3] supply the partial-

observation refinement sketched in §9. Schneider [4] shows EM-enforceable policies are safety properties, and denies the converse (§4).

The alphabet split is not ours. Closest to our setting is Basin, Jugé, Klaedtke and Zălinescu [8], who refine Schneider by distinguishing system actions an enforcement mechanism *controls* from those it merely *observes*, and give necessary and sufficient conditions for enforceability with respect to that distinction. Their (U, O)-safety condition, in closed form $\mathrm{cl}(\mathrm{pre}^*(P \cap U) \cdot O^*) \cap U \subseteq P$, specializes on a universal plant and regular K to exactly the criterion in our box; they identify Schneider's setting as the $O = \emptyset$ case, as we do. We claim no priority for the split, for the characterization, or for the observation that Schneider is its degenerate case. What Basin et al. leave explicitly open is the identification itself: they note that "the Ramadge–Wonham framework ... has several similarities with our setting" and close with "it remains to be seen whether and how the domain of policy enforcement can benefit from the Ramadge–Wonham framework and the results around it." This paper closes that question in one direction, and the closing is the first contribution. Naming the condition *controllability* is not a relabeling: it imports the supremal controllable sublanguage, the maximal-permissiveness certificate, and decades of modular, hierarchical and tool-supported synthesis, none of which the (U, O)-safety formulation provides — the $\sup \mathcal{C}(K) = \emptyset$ result of §8, which prices a thought-ban's best preventable approximation at nothing, has no counterpart in Basin et al. The second contribution is the instantiation: their canonical uncontrollable action is a clock tick, an event of the *environment*; ours are the generative model's own forward-pass steps, events of the regulated agent's own constitution — which is why the confident lie in a status report survives every widening of Σ_c: own the emission channel and the prohibition becomes controllable, but the gate condition is still a state of the agent the boundary never witnesses (boundary (vi), §9). A model whose uncontrollable events are clock ticks has no internal state of the regulated agent behind them and cannot state that corollary at all. Khoury and Hallé [11] generalize the binary split to a lattice of control levels; our two-valued split is the coarsening appropriate to a runtime that either owns a channel or does not.

Mechanism power is a different axis. The lineage descending from Schneider that classifies monitors by what they do to the trace is orthogonal to ours and should not be confused with it. Ligatti, Bauer and Walker [9] show that truncation, suppression, insertion and edit automata all *precisely* enforce exactly the safety properties, while an edit automaton *effectively* enforces any property by suppressing until a prefix is confirmed legal and then reinserting. Every one of those operations requires the mechanism to be interposed and able to withhold the action; edit automata are powerful precisely because they can *delay*, which is the capability an uncontrollable event denies. Our detect-and-compensate is therefore not a weaker member of their taxonomy but a category outside it — all four of their mechanisms are mechanisms over Σ_c — and they name our regime as an out-of-model limitation ("the monitor is unable to interpose itself between the application and the device"), which Basin et al. later formalize. Falcone, Fernandez and Mounier [10] ask a question whose title is nearly ours and answer it on the other axis: holding complete mediation fixed and equipping the monitor with unbounded finite memory, the enforceable properties are exactly the response properties of the safety–progress hierarchy. Their refinement moves *up* the hierarchy by adding buffering power; ours stays at prefix-closed safety and removes mediation. Neither contains the other, and the joint statement — safety *and* controllable — is what a deployed agent runtime must satisfy.

The word "regimentation" is not ours either. In normative multi-agent systems it is an established term of art for precisely the distinction this paper draws: regimentation makes violation impossible by design, enforcement detects violations and responds with sanctions [12]. That community has long observed both halves of our theorem informally — that regimenta-

tion operates by mediating access to resources and the communication channel, and that "the regimentation of all actions is often difficult or impossible" [13] — and has drawn our alphabet split as a design taxonomy, separating regimentation of an agent's *mental states* (possible only when the agent is a white box) from regimentation of its *actions* (the black-box case) [14]. An LLM agent is the black-box case, and Σ_u is the formal content of its black-boxness. One paper in this community goes further than observing the distinction informally: Dastani, Sardina and Yazdanpanah [16] formalize regimentability for a norm as supervisory-control-theoretic controllability, proved by the same Ramadge–Wonham theorem we import, though they present it as an unremarked corollary and do not pursue enforceability beyond regimentation, synthesis, or classification. Our contribution relative to this literature is to replace "often difficult or impossible" with a decidable criterion, a synthesis procedure, and a machine-checked classification of a deployed governance corpus.

Adjacent and contemporaneous. Ray [15] asks what a certified runtime gate can enforce for tool-using agents and answers on a third axis: safety plus decidability of good-prefix membership relative to an oracle interface, with a fallible judge and a statistical certificate. His gate sees and decides on every proposed action — $\Sigma_u = \emptyset$ in our terms — so his boundary and ours are complements, not competitors: he asks what a gate can *represent and decide*, we ask which events a gate can *refuse at all*. The reference-monitor lineage — Anderson's complete-mediation obligation [5], Rushby's kernel-verification standard [6], and Goguen–Meseguer's policy-as-noninterference framing [7] — supplies the architecture our Σ_c presumes and the companion theory Paper 4 builds on. Closest in spirit on the synthesis side is the *shielding* line: Bloem et al. [17] synthesize a reactive shield that sits between an untrusted component and its environment and corrects its outputs at runtime, and Alshiekh et al. [18] apply the same construction to a reinforcement learner, whose policy the shield cannot inspect and does not try to. A shield is a supervisor in our sense — it owns the action channel and nothing inside the learner — so that literature is an independent instance of our Σ_c/Σ_u split arrived at from reactive synthesis rather than from discrete-event control; what it does not ask is which events *admit* a shield, which is our question.

The search trail, and its gap. We swept four literatures by hand — supervisory control of discrete-event systems, runtime verification and enforcement, normative multi-agent systems, and the reference-monitor tradition — and the citations above are what that sweep returned. It was not a systematic search: there is no protocol, no enumerated venue list, no screening record, so "not found" here is not "proven nonexistent," and the two contribution claims of this section should be read with that discount applied. The adjacent fields most likely to hold a prior statement of our theorem and least covered by the sweep are safe reinforcement learning beyond the shielding papers above, LLM guardrail and inference-time filtering work, and the operating-systems capability literature. Boundary (vii) names the single outstanding item we already know about.

11 Honest boundary

What this theorem does NOT say. (i) Failure of exact realization does not rule out every safe restriction. Detection, audit, and remedies require their own evidence and response assumptions; they are not consequences of a failed controllability test. (ii) It does not certify any implementation: the theorem's Σ_c is the set of events *actually* mediated, and every unenumerated side channel (raw sockets, DNS, inherited file descriptors, /proc, git credential helpers, package managers, clipboard, provider callbacks) silently moves events from Σ_c to Σ_u and shrinks the regimentable set behind the operator's back — complete

mediation of the claimed alphabet is an architectural obligation (Anderson; Rushby) that a red-team, not this proof, discharges. (iii) The classification is relative to the alphabet split: a runtime that gates model steps (e.g. token-level filtering with the model inside the boundary) changes Σ_u and re-grades the table — the classification is relative to the declared architecture, not one fixed verdict. (iv) The full-observation assumption is critical: policies whose triggers are unwitnessed internal state fall to the Lin–Wonham refinement (§9) even when their prohibitions are controllable. Items (iii) and (iv) are the two axes of Figure 3: (iii) is the horizontal move, (iv) the vertical one, and a policy must make both trips to become preventable. (v) The checker verifies the stated policy automata over the stated alphabet [internal, `b3_controllability.py`]; it is a faithful implementation of the test, not a mechanized proof of the theorem — an Isabelle/Coq formalization remains the submission gate. (vi) A coarse event alphabet lacks a truth predicate for arbitrary reports. A restricted report over registered facts and checked receipts can have an evaluable output guard; this paper supplies no complete oracle for unrestricted prose and proves no impossibility for every richer fact model. (vii) Dastani, Sardina and Yazdanpanah's *Norm Enforcement as Supervisory Control* [16], obtained and read in full: their Theorem 1 proves a norm is "regimentable" iff a regimenting supervisor exists, by directly importing the same Ramadge–Wonham controllability theorem we do, in normative-multi-agent-systems vocabulary rather than ours. They do not claim the identification as a contribution — their stated novelty is extending it to sanction- and repair-based enforcement — and they do not address enforceability more generally, synthesis, or classification of a deployed corpus. This narrows but does not retract the priority claim in §10: the identification itself is folklore enough that two independent communities (Basin et al.'s security-policy line and this normative-systems line) import it in passing before this paper; what neither prior source gives is the synthesis procedure or the $\sup \mathcal{C}(K) = \emptyset$ result. The theorem and the classification stand regardless, since neither depends on being first.

Reproducibility. The policy table and the compound-case verdict regenerate deterministically from `skills/harbor-results/scripts/b3_controllability.py` (pure Python, no dependencies, no randomness); Figures 1 and 2 regenerate from `docs/harbor-research/figures/src/r5_figures.py` (the relation map with the script's third row dropped, since the regime diagram's right panel already carries it), and Figure 3 is drawn from those same checker verdicts plus the worked policy of §9. Every number and verdict in this paper is tagged [verified] (externally recomputable from the cited literature or by hand) or [internal] (regenerates from the named script).

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